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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not detected.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
not detected.
Subject Demographics
Age: not detected.
Weight: not detected.
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
Each strain was prepared in triplicate.
Number: Each strain was prepared in triplicate.
Data Information

Identifiers: Cloning, protein expression, and purification of Rv3705c and MSMEG_6251 in vitro The gene encoding full-length Rv3705c mtTiME, National Center for Biotechnology Information (NCBI) gene ID: 885229 UniProt accession no. I6XI06 was polymerase chain reaction (PCR)-amplified from M. tuberculosis H37Rv DNA library.	I6XI06
Identifiers: The gene encoding full-length MSMEG_6251 (msTiME, NCBI gene ID: 4530991 UniProt accession no. A0R5N2) was PCR-amplified from M. smegmatis mc2155 DNA library.	A0R5N2

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
For Western blot , the following primary antibodies were used: polyclonal anti-msTiME antibody (rabbit) , which was generated by immunization of rabbits with the purified His-tagged MSMEG_6251-SD fusion protein , in collaboration with the Institute of Genetics and Development , Chinese Academy of Sciences (1:2500 dilution) monoclonal anti-Antigen85 (mouse) from Santa Cruz Biotechnology (sc-57611 1:200 dilution) monoclonal anti-adenosine triphosphate (ATP) synthase b (AtpD) (mouse) from Invitrogen (A-21351 1:500 dilution) and polyclonal anti-GroEL (rabbit) from Abcam (ab90522 1:1000	anti-msTiME		
	anti-Antigen85		
	sc-57611		
	anti-adenosine triphosphate (ATP) synthase b (AtpD		
	A-21351		
	anti-GroEL		
Secondary antibodies used were horseradish peroxidase (HRP)-conjugated goat anti-rabbit immunoglobulin G (IgG) antibody from Promega (W401B 1:2500 dilution) and HRP-conjugated goat anti-mouse IgG antibody from Abcam (ab6789 1:1000	anti-rabbit immunoglobulin G (IgG		
	W401B		
	anti-mouse IgG		
	ab6789		
Experimental Models: Organisms/Strains			
Construction of msmeg_6251 knockout strain M. smegmatis mc2155 msmeg_6251 knockout strain was constructed by allelic recombination .	M. smegmatis mc2155 msmeg_6251 knockout		
Recombinant DNA			
Plasmids pJV53-green fluorescent protein (GFP) and PUC-Hyg (provided by Y . -C . Sun , MOH Key Laboratory of Systems Biology of Pathogens , Institute of Pathogen Biology , and Center for Tuberculosis Research , Chinese Academy of Medical Sciences and Peking Union Medical College , Beijing , P . R . China) were used for deletion of M. smegmatis msmeg_6251 gene (55) .	pJV53-green		
A 540-base pair (bp) DNA fragment upstream of msmeg_6251 (P1) and a 569-bp DNA fragment downstream of msmeg_6251 (P2) were amplified	PUC-Hyg		

by PCR and cloned into the PUC-Hyg vector to generate an integrating PUC-Hyg-P1-P2 plasmid (fig .			
The competent cells were electrotransformed with PUC-Hyg-P1-P2 plasmid to recombine with msmeg_6251 gene (fig .	PUC-Hyg-P1-P2		
GFP-negative colonies (loss of pJV53-GFP) were picked up for PCR (fig .	pJV53-GFP		
The pMV361D plasmid was derived from the pMV361 plasmid (62) by deleting ATG bases at the hsp60 sequence start site, in which its sequence was used to promote expression of the foreign antigen gene (62).	pMV361		
The genes of msmeg_6251, msmeg_6251 G33R-G50R-A91R, or msmeg_6251 N147R-F148R were cloned into the pMV361D vector.	pMV361D		
Software and Algorithms			
All of the crystallographic calculation was carried out with CCP4 package (43) , and the model building and correction were carried out in Coot (44) .	CCP4		Suggestion: CCP4 (RRID:SCR_007255)(link)
	Coot		Suggestion: Coot (RRID:SCR_014222)(link)
A total of 276,355 particles were picked automatically by RELION 3.0 (49) from 1335 micrographs , then extracted in 256 x 256 square pixels , and 2x binned to 128 x 128 square pixels (pixel size , 2.14 A per pixel) to speed up further data processing .	RELION		Suggestion: RELION (RRID:SCR_016274)(link)
Visualization of the EM density maps was made by UCSF ChimeraX (52) , and EM raw images were displayed by ImageJ (53) .	ImageJ		Suggestion: ImageJ, RRID:SCR_003070(link)
The absolute molecular weight was calculated using ASTRA 7 software (Wyatt Technology , Santa Barbara , CA , USA) based on a dndc value of 0.185 mlg (dn is the change in refractive index and dc is the change in concentration) .	ASTRA		Suggestion: ASTRA (RRID:SCR_016255)(link)
The raw data from Q Exactive were analyzed with Proteome Discovery 2.2.0.388 using Sequest HT search engine for protein identification and Percolator for false discovery rate analysis.	Percolator		
The UniProt M. smegmatis protein database (updated on March 2019) was used for searching the data.	UniProt		Suggestion: UniProt (RRID:SCR_002380)(link)

Differential expression analysis of two samples was performed using DESeq2.	DESeq2		Suggestion: DESeq, RRID:SCR_000154(link)
TopGO was used for GO (gene ontology) enrichment analysis.	TopGO		
ClusterProfiler was used for the classification and enrichment analysis of KEGG (Kyoto Encyclopedia of Genes and Genomes) pathway and COG (Clusters of Orthologous Groups).	ClusterProfiler		
	KEGG		

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq